Customer Segmentation Using Machine Learning

Arif Iftekhar Khan, 22201625 , Abdullah Farhan, 24241107

***Abstract*—This project explores a machine learning approach to predict customer segment labels (A, B, C, D) based on demographic and behavioral features. We utilize a dataset of 8,068 customers, containing attributes such as age, marital status, profession, spending score, and more. The data is first analyzed for patterns and then used to train classification models. We experiment with logistic regression, decision tree, and neural network algorithms. After data preprocessing and model tuning, the best model achieves about 50% accuracy . We examine the model's performance through confusion matrices and ROC curves, finding it can identify certain segments (notably segment D) with reasonable success, though some segments remain harder to distinguish. Key factors influencing the predictions include Age, Profession, Spending Score, and Marital Status. Overall, the model provides insight into segment characteristics and offers a modest predictive capability, highlighting both the potential and challenges of customer segmentation using the given features**

I. Introduction

Customer segmentation is a fundamental concept in marketing and retail, where customers are divided into distinct groups (segments) based on their characteristics or behaviors. Effective segmentation allows companies to tailor marketing strategies and services to different customer groups, improving customer satisfaction and business outcomes. Traditionally, determining a customer's segment might involve manual analysis or domain expertise. In this project, we aim to automate this process by using machine learning to predict the segment (labeled A, B, C, or D) of a customer from various personal and lifestyle attributes.

The goal is to develop a predictive model that can take inputs like a customer's age, marital status, education level, profession, spending habits, etc., and output the likely segment label. Such a model could assist businesses in classifying new customers in real-time, enabling targeted marketing (for example, offering different products to different segments) and personalized customer experiences. We approach this task as a multi-class classification problem with four classes (A, B, C, D).

II. Data Description

We use a structured dataset containing information on 8,068 customers, each with several features that potentially relate to their segment. Each customer is labeled with a segment (A, B, C, or D) which is the target variable for prediction. The dataset includes the following features for each customer:

Numerical Features:

- ID – A unique identification number for the customer (not used as a feature in modeling).
- Age – Age of the customer (in years).
- Work_Experience – Years of work experience the customer has.
- Family_Size – Number of family members the customer has (possibly including themselves).

Categorical Features:

- Gender – Gender of the customer (Male or Female).
- Ever_Married – Marital status (Yes if ever married, No if not).
- Graduated – Educational status (Yes if the customer is a graduate, No otherwise).
- Profession – Profession of the customer (e.g., Engineer, Doctor, Artist, Healthcare, etc.).
- Spending_Score – A self-reported or calculated spending score (Low, Average, or High), indicating the customer's spending habit or potential.
- Var_1 – An anonymized categorical variable with values like Cat_1, Cat_2, ..., Cat_7 (the exact meaning is not specified, but it may represent some grouped category or survey data).
- Segmentation – The target category indicating the customer's segment (A, B, C, or D).

III. Data Analysis

The dataset is fairly balanced among the four segments, with each segment represented by roughly 1,800–2,300 customers. Specifically, segment D has the most instances (approximately 2,268 customers, ~28% of the data) and segment B has the fewest (~1,858 customers, ~23%). Segments A and C each have about 1,970 customers (~24% each). This class distribution is shown in Fig 1. Balanced class counts are beneficial as they reduce the risk of model bias toward a majority class.

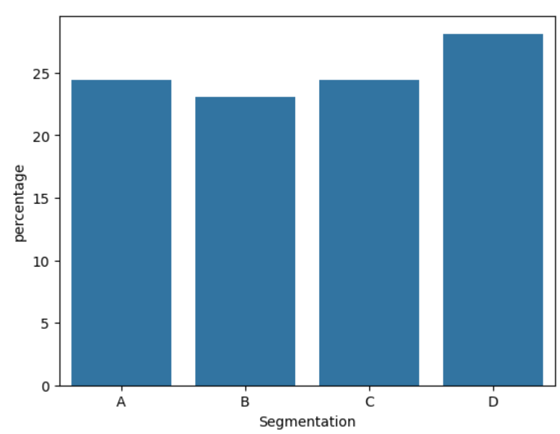

Fig. 1. Bar chart of the number of customers in each segment (A, B, C, D). The classes are relatively balanced in size, with each segment comprising roughly 23–28% of the 8,068 customers.

In addition to overall class balance, exploratory data analysis revealed notable differences in customer attributes across segments. For example, age varies significantly by segment. Figure 2 illustrates the age distribution for each segment using boxplots. We observe that segment D customers tend to be much younger on average than those in other segments. The median age in segment D is around 30 years, whereas segments A, B, and C have median ages around the mid-40s or higher. Segment D's age range is also more concentrated toward younger ages, while segments B and C include a broader range of older customers (with some up to around 80 years old). This suggests that segment D might represent a younger demographic group, whereas segments B and C correspond to older demographics. Segment A appears intermediate, but slightly younger than B and C. Such age differences indicate that age could be a strong predictor for distinguishing segment D from the others.

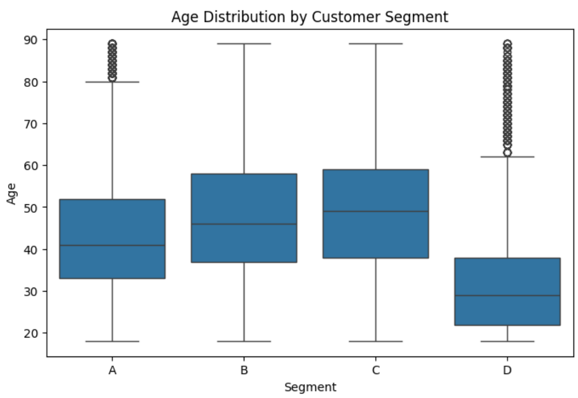

Fig. 2. Boxplot showing the age distribution for each customer segment (A, B, C, D).

In Fig, 2. segment D customers are visibly younger, with a median age around 30 and a lower overall age range, compared to segments A, B, and C whose median ages are roughly 45–50 and with many customers in older age brackets. This indicates age is a distinguishing factor, especially separating the predominantly younger segment D from the older segments B and C.

IV. Data Preprocessing

Dropped Unnecessary Fields: We removed the unique ID column since it does not carry predictive information for segmentation.

Handled Missing Values: For each feature with missing data, we filled in reasonable defaults and dropped some rows, For numeric fields like Work_Experience and Family_Size, we dropped rows with missing values. For categorical fields (e.g. Ever_Married, Graduated, Profession, Var_1), we filled missing values with the mode (most frequent value) of that column. For instance, missing Ever_Married entries were filled with "Yes" (since the majority $\tilde{5}8\%$ were married), and missing Profession with "Artist" (the most common profession). This simple imputation ensures no blanks remain, since most modeling algorithms cannot handle missing inputs and require complete data .

Encoding Categorical Variables: Because many models require numeric inputs, we converted categorical features into numerical form. We applied Label encoding for multi-category features like Profession (e.g. creating binary columns for Engineer, Doctor, etc.) and Spending_Score (Low/ Average/High), Ever_Married, Graduated and Gender.).

Feature Scaling: We standardized the numeric features (Age, Work_Experience, Family_Size) using MinMax Scaler. This step puts these features on a comparable scale. Although tree based models don't require feature scaling, methods like K-Nearest Neighbors and Neural Networks work better when features are normalized. For example, Age originally ranged up to $\tilde{9}0$, which without scaling could dominate distance calculations in KNN or affect neural network training..

Train-Test Split: Finally, we split the processed data into a training set (70%) and a test set (30%).

V. Model Training

We trained three different classification models on the prepared dataset: a Logistic Regression, a Decision Tree, and a Neural Network. We chose these to cover a range from simple linear to more complex non linear learners. All models were trained on the same training data and evaluated on the test set using accuracy (percentage of correct predictions) as the metric.

Logistic Regression*:* We used a multinomial logistic regression as a baseline. Logistic regression is a linear model that makes predictions based on a weighted sum of the features (after one-hot encoding and scaling). It's fast to train and provides a good reference performance.

Decision Tree*:* We trained a decision tree classifier, which splits the data based on feature values to create a flowchart-like model of decisions. Decision trees can capture non-linear relationships and interactions between features. To avoid overfitting the tree to the training data, we limited the tree's maximum depth. We initially set max_depth = 5 (allowing at most 5 levels of splits) so that the tree would focus on the most important splits first. A shallow tree is also easier to interpret, which helps in understanding the decision logic.

Neural Network: We included a neural network because it's a powerful non-linear model capable of automatically learning complex feature combinations. Given our data contains many categorical features and potential interactions (like Age and Marital Status jointly indicating a segment), a neural net can capture those subtle patterns that linear models might miss.

All models were implemented using scikit-learn within a unified pipeline. The pipeline ensured that the same preprocessing was applied consistently to each model, and there was no data leakage from the test set during training or tuning. We used appropriate random seeds for reproducibility of results

VI. Model Evaluation

Accuracy: the percentage of correct predictions on the test set. This gives an overall measure of performance. However, accuracy alone can be misleading if classes were highly imbalanced (not the case here) or if we care about specific classes more. Still, it's a primary metric for overall success

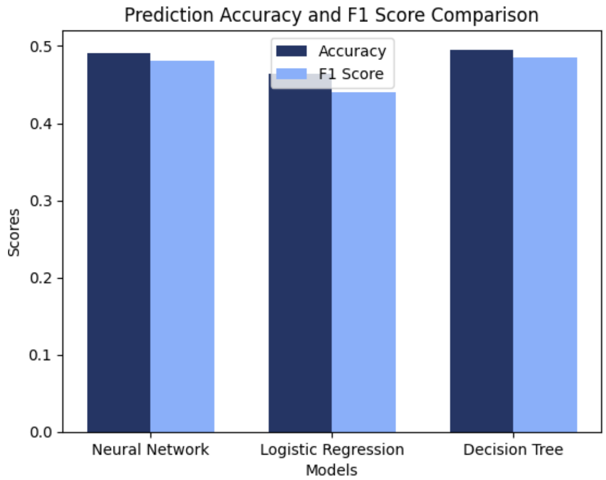

Fig. 3. This bar chart compares the overall prediction accuracy and F1 scores of the three models.

The Decision Tree and Neural Network models perform similarly, each achieving close to 49–50% accuracy and F1 scores, indicating balanced performance. Logistic Regression slightly lags behind, with both accuracy and F1 score near 46%. The close values of accuracy and F1 score suggest that precision and recall are fairly balanced for all models..

Confusion Matrix**:** we computed the confusion matrix for the test results, which shows counts of correct and incorrect predictions for each class. This allowed us to see which segments are often confused with each other (e.g., are A and B frequently mixed up by the model? Does the model reliably identify D?).

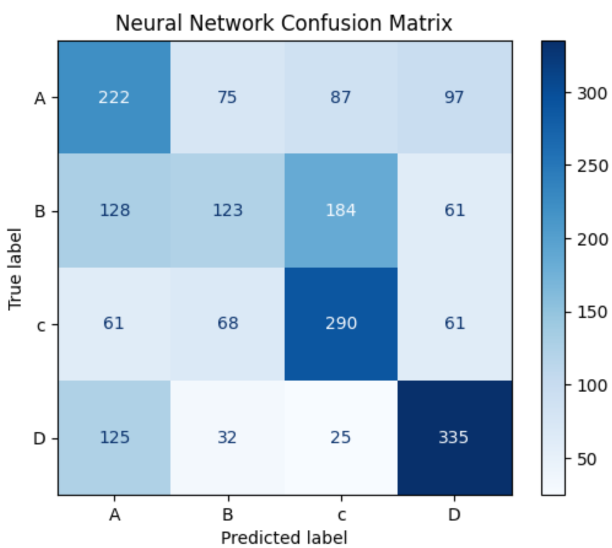

Fig. 4. Confusion matrix of Neural network.

This confusion matrix shows that the neural network performs best on segment D, correctly predicting 335 out of the actual segment D customers, achieving around 77% recall. Segment C is also predicted well, with 290 correct classifications. However, there is notable confusion between segments A and B, where many customers are misclassified as each other. This suggests that the model identifies segments D and C effectively, but finds it challenging to distinguish between the overlapping characteristics of segments A and B

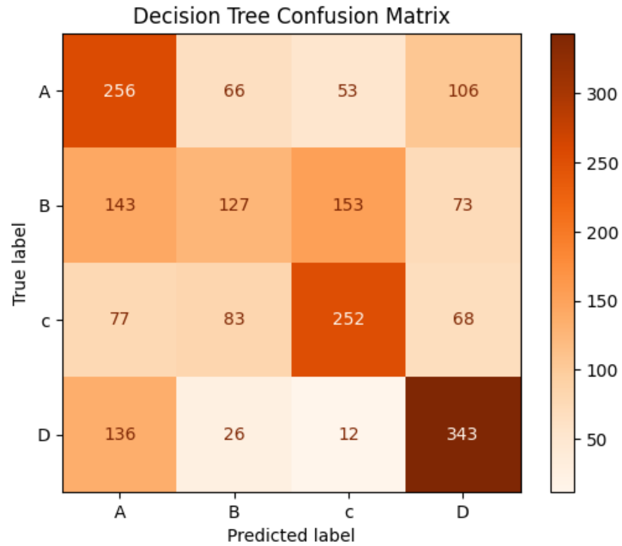

Fig. 5. Confusion matrix of Decision Tree.

In Fig, 2. segment D *Figure 5: This decision tree confusion matrix shows that the model performs strongest on segment D, correctly identifying 343 customers. It shows decent accuracy for segments A and C, but struggles with segment B, which is often misclassified as either A or C. The confusion across segments indicates that while the decision tree captures certain structure in the data, it lacks the depth or flexibility to fully separate the segments like B.*

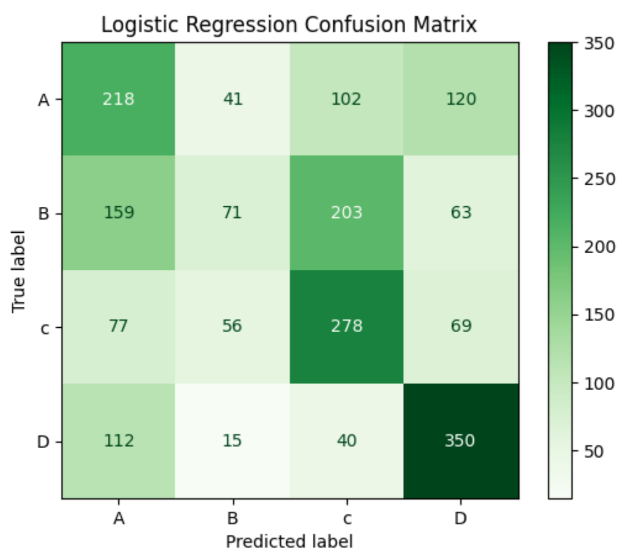

Fig. 6. Confusion matrix of Logistic Regression.

Figure 6. matrix illustrates that logistic regression predicts segment D (350 correct) and segment C (278 correct) reasonably well. However, segment A shows high misclassification, with many customers being predicted as segment C or D.

Segment B is also often confused with segment C. These results reflect the limitations of linear boundaries, as logistic regression lacks the complexity to separate more subtly distinct groups like A and B.

ROC Curves and AUC: Since this is a multi-class problem, we generated one-vs-rest ROC curves for each class. The ROC curve plots the true positive rate against the false positive rate at various threshold settings, and the area under this curve (AUC) indicates how well the model separates that class from the others. We computed the AUC for each segment

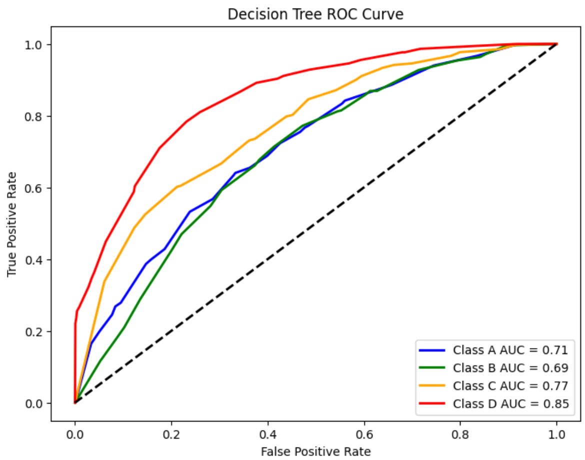

Fig. 7. Roc Curve of Decision tree.

In fig. 7. decision tree show that it performs best on segment D (AUC = 0.85) and reasonably well on C (AUC = 0.77). A and B show lower AUCs (0.71 and 0.69), confirming earlier confusion seen in the confusion matrix. The model is consistent in ranking D well, with moderate results on other classes. Average AUC is 0.755.

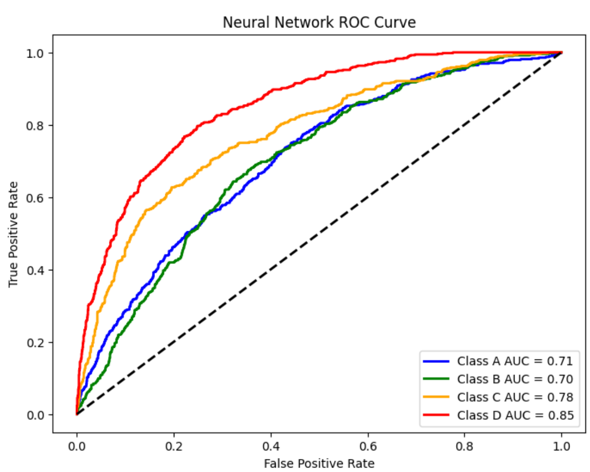

Fig. 8. Roc Curve of Neural network.

In fig. 8. the neural network achieves the highest AUC for class D (0.85), indicating strong separation of this segment. Class C also performs well (AUC = 0.78), while classes A and B are less distinguishable (AUCs around 0.71 and 0.70 respectively). This shows the model can confidently rank segment D customers but has more difficulty with A and B. Average AUC is 0.759.

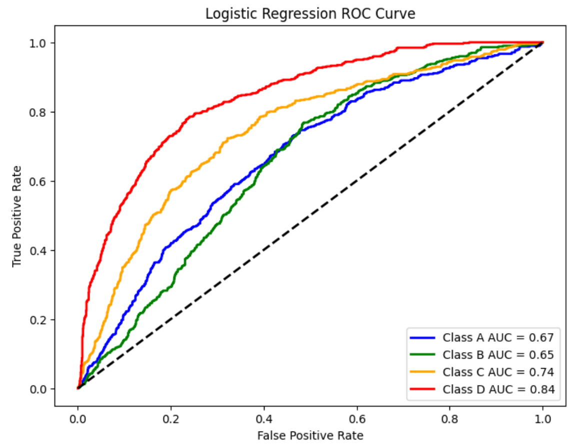

Fig. 9. Roc Curve of Logistic Regression.

Figure 9. shows that Logistic regression gives its best AUC for segment D (0.84), with C following at 0.74. It performs less effectively on classes A and B (AUCs of 0.67 and 0.65), highlighting difficulty in cleanly separating these groups. This reflects the model's simpler, linear nature. Average AUC is 0.726.

Performance metrics:

Class	Precision	Score	F1-Score	Support
A	0.41	0.46	0.44	481
B	0.41	0.25	0.31	496
C	0.49	0.60	0.54	480
D	0.60	0.65	0.63	517
Accuracy			0.49	1974
Macro avg	0.48	0.49	0.48	1974
Weighted avg	0.48	0.49	0.48	1974

TABLE I
PERFORMANCE METRICS OF NEURAL NETWORK.

Class	Precision	Score	F1-Score	Support
A	0.39	0.45	0.42	481
B	0.39	0.14	0.21	496
C	0.45	0.58	0.50	480
D	0.58	0.68	0.63	517
Accuracy			0.46	1974
Macro avg	0.45	0.46	0.44	1974
Weighted avg	0.45	0.46	0.44	1974

TABLE II
PERFORMANCE METRICS OF LOGISTIC REGRESSION.

Class	Precision	Score	F1-Score	Support
A	0.42	0.53	0.47	481
B	0.42	0.26	0.32	496
C	0.54	0.53	0.53	480
D	0.58	0.66	0.62	517
Accuracy			0.50	1974
Macro avg	0.49	0.50	0.48	1974
Weighted avg	0.49	0.50	0.49	1974

TABLE III
PERFORMANCE METRICS OF DECISION TREE.

From the classification reports given in tabel 1, tabel 2 and tabel 3 we can compare the precision, recall, and F1-scores of the Neural Network, Logistic Regression, and Decision Tree models across all four customer segments. The Neural Network performs moderately well overall with 49% accuracy, showing strong F1-scores for Segment D (0.63) and Segment C (0.54), but struggles significantly with Segment B (0.31). Logistic Regression has the lowest overall accuracy (46%) and especially poor recall for Segment B (14%), despite reasonable performance on Segment D. In contrast, the Decision Tree model achieves the highest overall accuracy (50%) and the best weighted average F1-score (0.50), with balanced performance across segments—especially strong results for Segment C (F1 = 0.53) and Segment D (F1 = 0.62). All models consistently perform best on Segment D and weakest on Segment B, highlighting challenges in distinguishing overlapping class features.

VII. Conclusion

Based on our evaluation, the Decision Tree emerged as the best-performing model among the three, achieving the highest overall accuracy (50%) and weighted average F1-score (0.50). While the Neural Network closely followed with balanced metrics and strong performance on Segment D, the Decision Tree showed more consistent results across multiple segments. Logistic Regression, although simple and interpretable, lagged behind in both accuracy (46%) and recall—especially for Segment B, where it struggled significantly.

The overall moderate accuracy of all models (ranging from 46% to 50%) reflects the inherent difficulty of the segmentation task using the available features. While some segments—particularly Segment D—were more distinct and thus easier for models to classify (thanks to unique combinations like younger age and low spending), other segments like A and B had overlapping feature distributions. This overlap likely caused frequent misclassifications and reduced model confidence, especially in linear models like Logistic Regression.

These results suggest that the current feature set provides partial but not complete discriminative power for cleanly separating all segments. Age, Profession, Spending Score, and Marital Status were the most influential variables, and future improvements could involve introducing additional customer behavior data—such as income level, purchase history, or digital engagement metrics. Nonetheless, this project demonstrates the practical potential and limitations of using machine learning for customer segmentation.